Total Number of Pages 3 Roll No.....

B.Tech. Electrical Engineering **5th SEMESTER**

END SEMESTER EXAMINATION *(Nov. 2025)*

EE-301-Power Electronics

Time: 3:00 Hours ***Maximum Marks:40***

Note: Assume suitable missing data, if any.
Attempt all parts of a guest Q at one place (*Marks may not be awarded otherwise*)
Q.1 is COMPULSORY attempt other four from the rest

1 **Draw *neat* waveforms/ characteristics (*free hand*) depicting requisite details**

(a) Through different I-V characteristics of Thyristors depict the Turn-On issue while they are connected in series, and looking for voltage equalization.

(b) Through reverse recovery characteristics of power diodes (UF/Fast/Rectifier Grade) depiction ΔQ_{RR}, and different time(s) describing the snappiness factor

(c) Switching characteristics during Turn-On of the power MOSFET feeding RL load. Label the diagrams (v_G, i_D and v_{DS}) appropriately.

CO 1,2,5

(d) Current & Voltage waveforms both on AC and DC sides of a single phase full thyristorised AC-DC converter feeding a large inductive load with firing angle of 90°. Quantify the involved power transfer to the load.

(e) Input and Output side current waveform (both on AC and DC sides) of diode rectifier feeding resistive load without and with filter capacitor.

(f) Current and voltage waveform of single-phase AC voltage regulator feeding to purely inductive load (Thyristor Controlled Reactor) for α=150°. Draw the voltage waveform across the switches too.

(g) Waveform of inductor voltage, inductor current and capacitor current of boost convertor with reference to gating pulses.

(h) Differentiate the voltage swing at the output of the single-phase inverter operating with Bipolar and Unipolar sinusoidal pulse width modulation through the waveforms. Identify the spectrum of harmonics. 1x8

2 [a] Derive the expression form the fundamental for the R & C parameters of a RCD turn-off snubber. Design a turn-off RCD snubber for a transistor switch operating at 100kHz with a duty ratio of 50% and transistor turn-off time of 0.1μs. Operating voltage is 200V and load current is 10A. Use criterion that switch voltage reaches V_s when switch current reaches zero. Determine overall turn-off losses with and without snubber Assume, a time equal to 5-time constants is necessary to discharge the snubber capacitor when the switch is turned ON next. 4

CO 2-3

[b] Derive the design equations for the LC output filter for the single phase diode bridge rectifier. The rectifier is fed from 230V, 50Hz supply and feed to a resistive

load of 5 Ω. An LC filter (L=5mH and C=10,000μF) is deployed on the output terminals. Comment on conduction mode & compute output voltage ripple. 4

3 [a] The battery charger realized with a three-phase thyristor bridge rectifier fed from 415V, 50 Hz AC source with series RL impedance (R=20Ω; L=200μH) on DC side connected in tandem with 8 units of 12V 62Ah lead acid battery packs. The nominal cell voltage is 2V, whereas, under deep discharge, the cell voltage reaches 1.8V and end of the constant current charging is marked by 2.267V per cell. Determine the change in the angle of triggering for the thyristors so the load current is held constant during charging of the battery pack. For the minimum battery voltage, the angle of triggering (α) for the thyristors is kept at α = 45° to realize charging @C/4. CO 2,3,4 4

[b] Derive the fourier coefficients a_n & b_n of the DC side voltage of a thyristorized full converter feeding load having large inductance. Compute the power absorbed by the load and *pf* when firing delay is α=60°. The connected load is R=10Ω and L=100mH, and source side voltage is 230V, 50Hz. Consider only 2nd harmonic on DC side current for the computation of power and *pf*. 4

4 [a] Derive various relations and mathematical expression necessary for a design of buck converter assuming continuous mode of operations. The buck converter operates with the following parameters: V_{in}=80 V, R_o=18 Ω, P_o=100W, L=0.4 mH, and f_s=150 kHz. Determine the mode of operation, and estimate the range R_o for the converter to remain in the CCM. CO 2,3,4 4

[b] Deduce various relations and mathematical expression necessary for a design of boost converter assuming continuous mode of operations. If the duty ratio is adjusted to regulate the output voltage at 48 V. The input voltage varies in wide range from 12-36V. The maximum power output is120W. The converter has to operate in continuous conduction mode with switching frequency of 50kHz. 4

5 [a] An AC/AC voltage regulator operating with two thyristors in anti-parallel connection and is fed from 220 V, 50 Hz supply, and in turn feed a heater load R_L = 2.2 Ω. The voltage drop across thyristor while conducting V_{T1} = 1 V and the resistance while conducting r_d = 5 mΩ. If the required power of the load (heater) is P_L = 11 kW, determine: (*Hint- attempt substituting the value of α to be standardised values, viz, 0°,30°,60°,90°,120°,150° and 180°*)

(a) angle of conduction of the thyristors,
(b) load power factor,
(c) power of dissipation in one of the thyristors, 4

[b] Discuss in brief with circuit diagrams, waveforms of the single phase to single phase and three phase to single phase cycloconverter. Compare the operation of single phase to single phase cycloconverter with dual converter. CO 2- 5

OR

Discuss in brief details with waveforms the square and quasi square output voltages of a single phase bridge inverter. Derive the output voltage and current expressions of the inverter. Comment on the comparative harmonic components of the two schemes and support your claim with requisite derivation of expression on how certain harmonics are eliminated by selectively shaping the quasi square wave shape. 4

6 [a] Draw neat waveforms of 3-phase thyristorised inverter voltages (v_{ao}, v_{bo}, v_{co}), line-line voltages (v_{ab}, v_{bc}, v_{ca}) & load voltages (v_{an}, v_{bn}, v_{cn}) for 120° conduction. Derive the expression for the load voltages (v_{an}, v_{bn}, v_{cn}). Comment on the waveshapes and phase displacements of the fundamental components of different aforesaid voltages. CO 2-5 4

[b] The switches of a *1Ph* full bridge inverter are switched with bipolar SPWM. The DC bus is kept at 400V, terminal voltage of the inverter 223V AC and rms current output is 22A. Compute the fundamental rms current in the switches. 4

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V SEMESTER B.Tech[EE]

END TERM EXAMINATION NOV-2025

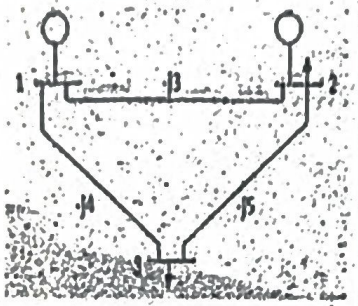
EE 303 POWER SYSTEM ANALYSIS

Time: 3:00 Hours Max. Marks: 40

Note: Answer any five questions. All questions carry equal marks. Assume suitable missing data, if any.

Q.1 a) Write about differences between Gauss-Seidel and Newton Raphson methods. Also explain algorithm for Decoupled method. [4][CO1][BTL1]

b) For a three bus system shown in fig., determine the voltages at buses 2 and 3 after 1st iteration using Gauss-Seidal method. Take the acceleration factor α=1.6. [4][CO1][BTL3]



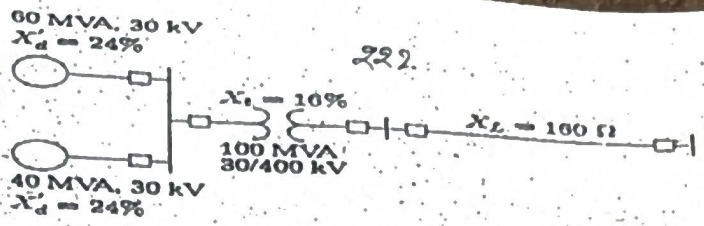
Bus No	Type	Generation (PU)		Load (PU)		Bus Voltage	
		P_G	Q_G	P_L	Q_L	V	δ
1	Slack	-	-	-	-	1.02	0
2	PQ	0.25	0.15	0.5	0.25	-	-
3	PQ	0	0	0.6	0.3	-	-

Q.2 a) Explain about generation allocation in power system including the effect of transmission line loss [4][CO2][BTL4]

b) In a two plant system, the entire load is located at plant 2, which is connected to plant 1 by a transmission line. If a load of 125 MW is transmitted from plant 1 to the load, there is a loss of 12.5 MW. Determine the generation schedule and the load demand if the cost of received power is Rs. 70 per MWh. Assume incremental costs of two plants as: $IC_1 = 0.25 P_1 + 40$ Rs/MWh & $IC_2 = 0.2 P_2 + 50$ Rs/MWh. [4][C2][BTL5]

Q.3 a) Explain how symmetrical faults are analysed? Differentiate between sub transient and transient reactance. [4][CO3][BTL4]

b)The system shown in fig is initially on no load with generators operating at their rated voltage with their emfs in phase. The rating of the generators, & transformers and their respective percent reactance are marked in fig. All resistances are neglected. The line impedance is j160 ohm. A 3 phase balanced fault occurs at the receiving end of the transmission line. Determine the short circuit current and short circuit MVA. [4][CO3][BTL3]



Q.4 a) Derive the relationship to determine the fault current for a line to line fault. Draw an equivalent network showing the interconnection of sequence networks to simulate L-L fault. [4][CO4][BTL1]

b) A three phase 10MVA, 11 kV synchronous generator with a solidly grounded neutral point supplies a feeder. The relevant impedance of the generator and feeder are as follows:

	Generator	Feeder
Positive sequence impedance Z_1	j1.2 ohm	j 1.0 ohm
Negative sequence impedance Z_2	j 0.9 ohm	j 1.0 ohm
Zero sequence impedance Z_0	j 0.4 ohm	j 3.0 ohm

A fault from one phase to ground occurs at the far end of the feeder. Determine the voltage to neutral of the faulty phase at the terminals of the generator. [4][CO4][BTL3]

Q.5 a)Explain the necessity of neutral grounding and what are the consequences of an ungrounded system. [4][CO2][BTL2]

b) The resolution of a set of three phase unbalanced voltages into symmetrical components gave the following results: $V_{a0} = 30\angle -30^\circ$ V, $V_{a1} = 450\angle 0^\circ$ V, V, $V_{a2} = 225\angle 40^\circ$ V. Current components are: $I_{a0} = 10\angle 190^\circ$ A, $I_{a1} = 6\angle 20^\circ$ A, $I_{a2} = 5\angle 50^\circ$ A . Determine the complex power represented by these voltages and currents by (a) symmetrical components (b) unbalanced phase components. [4][CO4][BTL3]

Q.6 a) Draw flow chart for load flow study using NR method. How does the method get modified to account for PV buses? [4][CO1][BTL1]

b) Derive the transformation matrix for converting Phase voltage to Sequence Voltage and vice versa. [4][CO4][BTL#]

Q.7 Write short notes on any two of the following.

a) Current limiting reactors

b) Z bus building algorithm

c) Jacobian matrix in power system [8][CO1,2,3][BTL1]

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FIFTH SEMESTER B.TECH(EE)

END TERM EXAMINATION NOV-2025

EE-305 MICROPROCESSORS AND MICROCONTROLLER APPLICATIONS

TIME:3HRS **MAX. MARKS:40**

NOTE: *ATTEMPT ANY FIVE QUESTIONS*

1(a)	Write an assembly language program to find positive numbers in a data array of 10 elements. The length of block is in memory location starting from 9100H. Store the result in memory location 9200H	4	[CO2] BTL 1,2
(b)	Describe the step-by-step execution of Booth's Algorithm(shift and add method) for multiplying (-6) and (+7) using 5-bit binary numbers. Show all intermediate values of the Accumulator (A), Multiplier (Q), Multiplicand (M), and Q-1 (Q minus one) at each iteration of the algorithm.	4	[CO2] BTL 1,2
2(a)	Write an assembly language program assuming microprocessor is completing RST7.5 interrupt request, check to see if RST6.5 is pending. If it is pending, enable RST6.5 without affecting any other interrupts otherwise return to the main program.	4	[CO3] BTL 3
(b)	Draw block diagram explaining 8085 interrupts and vector locations. Also explain the bit pattern for RIM and SIM instructions	4	[CO3] BTL2
3(a)	Explain control word of programmable interval Timer 8254. Discuss with timing diagram the six different modes in which the 8253/8254 Programmable Interval Timer can operate?	4	[CO3] BTL 3
(b)	Write an assembly language program to generate a square wave of 1 KHz on display device connected to SOD pin of 8085 microprocessor	4	[CO3] BTL 3
4(a)	Draw and explain a typical stepper motor interface with microprocessor. Write an ALP to rotate the stepper motor in clockwise direction. With 1ms time delay between each step.	4	[CO5] BTL 4,5
(b)	With the help of block diagram explain keyboard interfacing IC 8279.	4	[CO4] BTL 3
5(a)	Draw the functional block diagram of 8251 USART. Also, explain the operations of the transmitter and receiver section of 8251 USART.	4	[CO4] BTL 2,3

(b) With the help of architecture explain the working of 8051 microcontroller. 4[CO4] BTL 3

6(a) Draw 8085 microprocessor based interfacing circuit to interface ADC 0809 through 8255. Extend this circuit for measurement of temperature. Write the instruction to display the value of temperature to PORT B of 8255. 4[CO5] BTL 5,6

(b) Draw 8085 microprocessor based interfacing circuit to measure time period of sinusoidal voltage waveform. Write the instruction to display the value of time period at PORT A of 8255. 4[CO5] BTL 5,6

7(a) Design an I/O interfacing circuit using 8085 microprocessor such as to display a 8 bit hexadecimal number 02H on the common anode seven segment LED display having address as 01H 4[CO2] BTL 3,4

(b) Predict the output with justification 2[CO1] BTL 1,2

```
MVI B, 00H;
RLC;
JNC **;
RRC;
ADD A;
JNC **
INR B;
** HLT;
```

(c) Correct the following program predict its output with justification. 2[CO1] BTL 1,2

```
MXI H, 1400H;
MVI A, 56 H;
MOV M, A;
MOV D, 35H;
MVI E, 25H;
MOV A, D;
ADD M;
```

Total no. of Pages: 03 *Roll no.....*

5th and 7th SEMESTER

B.Tech.

END TERM EXAMINATION ***Nov-2025***

COURSE CODE:EE319 COURSE TITLE: Digital System Design

Time: 3:00 Hours ***Max. Marks: 50***

Note :
If necessary, make appropriate assumptions and approximations and state them.You must solve all the questions and their parts in the serial order

Q.1 Answer any SIX. [3X6=18]

(a) What is the function of a transmission gate? Why is it preferred over a single NMOS pass transistor? [CO#2][BTL1,2]

(b) Explain the concept of body effect in MOS transistors. How does it affect threshold voltage? [CO#1][BTL1,2]

(c) What are the main differences between static and dynamic power dissipation in CMOS circuits? [CO#2][BTL1,2]

(d) Define setup time and hold time of a flip-flop. [CO#3][BLT1]

(e) Sometimes the substrate is connected to a voltage called the substrate bias to alter the threshold of the nMOS transistors. If the threshold of an nMOS transistor is to be raised, should a positive or negative substrate bias be used? With proper justification. [CO#1][BLT3]

(f) Why TSPC latches are preferred as compare to Static D latches? [CO#3][BLT4]

(g) You are considering lowering VDD to try to save power in a static CMOS gate. You will also scale Vt proportionally to maintain performance. Will dynamic power consumption go up or down? Will static power consumption go up or down? [CO#3][BLT4]

Q2. Answer any TWO. [5X2=10]

(a) Sketch a 4X1 multiplexer. You are given four data signals D0, D1, D2, and D3, and two select signals, S0 and S1. How many transistors does each design require? [CO#2][BLT3,6]

a) Use only static CMOS logic gates.

b) Use a combination of logic gates and transmission gates.

(b) Design a 2 input XOR gate using transmission gate logic and write a Verilog module for the circuit designed. [CO#4][BLT6]

(c) Consider the nMOS transistor in a 0.6 um process with gate oxide thickness of 100 Å. The doping level is $N_A = 2 \times 10^{17} cm^{-3}$ and the nominal threshold voltage is 0.7 V. The body is tied to ground with a substrate contact. How much does the threshold change at room temperature if the source is at 4 V instead of 0? [CO#2][BL4]

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Q3. Design and explain the working of C^2MOS Flip-Flop. [5][CO#3][BLT1,2,6]

Q4. Ignore all capacitances at the "out" node: [5][CO#2][BLT4,6]

(a) In the high to low transition in Figure 1, which transistor would first turn from ON to OFF state?

(b) During the high to low transition in Figure 1. What are the regions of operation of transistors Mp1 – Mp3 and Mn1 – Mn3 for Vin = 0.7V (Vth = 0.3V)?

(c) What is the value of the current through PUN?

(d) During the low to high transition, which of Mn1 and Mn2 first turns ON?

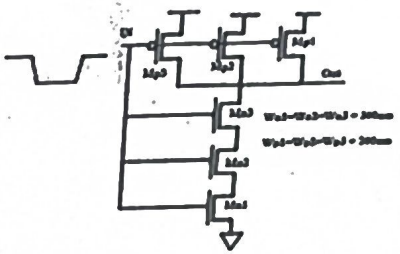


Figure 1

Q5. Write a 3 bit squarer circuit in Verilog code. [7] [CO#4][BLT6]

Q6. Determine the output: [5][CO#2][BLT4]

a) What logic equations do the following schematics implement in Figure 2?

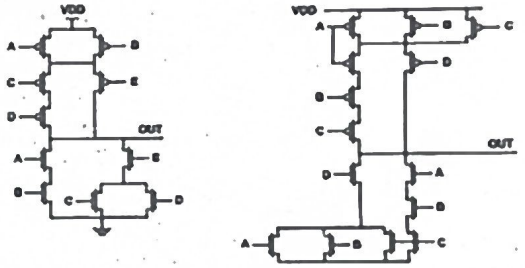


Figure 2

b) What is the output function of the following circuit show in Fig. 3?

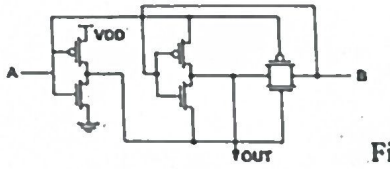


Figure 3

sets. (If there are multiple possible sets of principal components, you need to consider only one of them.)

ii. In which of the above cases will the first Principal Component capture a larger proportion of the variance in the data? Explain your answer.

iii. In which of the above cases will the first two principal components (combined) capture a larger proportion of the variance? Explain your answer.

Q.5 [Clustering] **[5 +5][CO5][BTL 3, 4]**

A. The k-means algorithm is designed to reduce the mean squared error between the cluster centers and their assigned points. Will increasing the value of k ever increase this error term? Why or why not?

B. Use K Means clustering to cluster the following data into two groups. Assume cluster centroid are C_1=2 and C_2=4. The distance function used is Euclidean distance.
Data: { 2, 4, 10, 12, 3, 20, 30, 11, 25 }

Q.6 [Application Based: Explainable AI] **[5 +5] [CO 5] [BTL 2, 5]**

A. Explain the difference between Transparency, Interpretability, and Explainability with suitable examples.

B. Describe LIME and discuss its strengths and limitations as a post-hoc explanation method.

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Total no. of Pages: 4 *Roll no.....*

V & VII SEMESTER

B.Tech

END TERM EXAMINATION ***Nov/Dec-2025***

EE323 FUNDAMENTALS OF MACHINE LEARNING

Time: 03:00 Hours ***Max. Marks: 50***

Note : All questions carry equal marks (10 marks).
Attempt "ANY 5" questions
Assume suitable missing data, if any.

Q.1

A. Write the correct answer and justify: [2 + 2 + 2] [CO 1,2] [BTL 1, 2]

i. Let X be normally distributed with mean = 14 and variance = 81, the value of b for which $E(X - b)$ is least, is _____.

ii. The angle between two vectors $\bar{A}_1 = [2\ \ 6\ \ 14]^T$, and $\bar{A}_2 = [-12\ \ 8\ \ 16]^T$ in radians is _____.

iii. For a random variable 'X' $(-\infty < x < \infty)$ following normal distribution, the mean is 100. If the probability is P = 0.39 for $x \geq 110$, then P $[90 \leq x \leq 110]$ is_____.

B. For each statement below, say whether it is true or false, and give a one or two sentence justification of your answer. **[2 + 2] [CO 2, 3] [BTL 2]**

i. As we increase the number of parameters in a model, both the bias and variance of our predictions should decrease since we can better fit the data.

ii. Machine learning models work automatically without the need for the user to manually set hyperparameters.

Q.2 [Logistic Regression + GD Optimization] [5 + 5(2 +3)] [CO 3, 4] [BTL 2, 3, 4]

A. Discuss why the **logistic function** is an appropriate hypothesis function for logistic regression, particularly in comparison to a linear hypothesis function. Also justify why the **Binary Cross-entropy Loss** is preferred over **MSE (Mean Squared Error)** for logistic regression despite MSE being used in linear regression.

B. Write short notes on Gradient Descent Algorithm? Explain with a pseudo code? Let us say that we are fitting a one-parameter model to the data, i.e., $y_n \approx \beta_0$. The average of $y_1, y_2, \ldots, y_N$ is 1. We start gradient descent at $\beta_0^{(0)} = 0$ and set the step-size to 0.5. What is the value of β_0 after **3 iterations**, i.e., the value of $\beta_0^{(3)}$?

Q.3 [Neural Networks] [3 + 3 + 4] [CO 4] [BTL 3, 4, 5]

Consider a single hidden unit (z) in a neural network. The unit receives (D) input features ($x_1, x_2, \ldots, x_D$), with corresponding weights ($w_1, w_2, \ldots, w_D$). A bias term is included using a dummy input ($x_0 = 1$) with weight (w_0).
The activation function of the unit is the **Leaky ReLU**, defined as:

$$LR(a) = \begin{cases} a, & a > 0 \\ 0.01a, & a \leq 0 \end{cases}$$

The pre-activation is, $a = \sum_{i=0}^{D} w_i x_i$, and the output of the hidden unit is, $z = LR(a)$.

(a) Draw a clearly labelled diagram of the hidden unit (z), showing:
- all input nodes ($x_0, x_1, \ldots, x_D$)
- all incoming weights ($w_0, w_1, \ldots, w_D$)
- the summation node computing ($a = w^T x$)
- the activation function node producing ($z = LR(a)$)

Write the expression for (z) explicitly in your diagram.

(b) During backpropagation, we wish to compute the gradient of the error(E) with respect to each weight:

$$\frac{\partial E}{\partial w_i}, \; i = 0, 1, \ldots, D$$

Using the chain rule and treating (z) as the intermediate variable, express $\frac{\partial E}{\partial w_i}$, as a product of two partial derivatives. Explain where the first derivative in your product comes from during backpropagation.

(c) To complete the chain-rule expression, we require the derivative of the Leaky ReLU activation. Compute $\frac{\partial LR(a)}{\partial a}$ for all values of a. Is this derivative undefined at any point? Justify your answer.

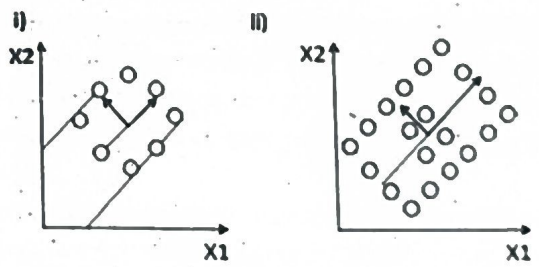
Q.4 [Dimensionality Reduction] [4 +6(2 + 2 + 2)] [CO 4, 5] [BTL 3, 4, 5]

A. Determine the Singular value decomposition (SVD) of the following matrix "A"

$$A = \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & 0 \end{bmatrix}$$

Also, comment on how the eigenvalues and singular values are related?

B. **Figure (i)** and **Figure (ii)** below show two different 2-dimensional data sets plotted on axes X_1 and X_2.



i. Write down the unit vectors corresponding to the **first** and **second Principal Components** (PC) for both of the above data

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V / VII SEMESTER

B.Tech. [EE]

END-SEM EXAMINATION DEC.-2025

EE325 INTERNET OF THINGS

Time: 3 Hour Max. Marks: 50

Note: Attempt any five questions. All questions carry equal marks

Q.1[a] Compare wireless sensor network versus IoT. [3] CO2 L2

[b] Discuss the IoT stack in detail with examples for each layer. [7] CO1, CO4 L2

Q.2[a] Explain the working of MQ-02 sensor and its interfacing with the Arduino microcontroller. How is MQ-05 different from MQ-02. [5] CO4 L2

[b] Explain the MQTT scheme: publish – subscribe pattern. Also explain the three levels QoS supported by MQTT. [5] CO2 L2

Q.3[a] Discuss the various types of CoAP – request – response layer messages. [4] CO2 CO5 L2

[b] Discuss the various types of transport protocols. [6] CO2 CO5 L3

Q.4[a] One of the PC on DTU network shows the IP address as 10.13.10.94. Which version of IP address is this? What should be its subnet mask? What is the class of this IP address and justify the use of this class of IP address for DTU network. [6] CO3 L2 L3

[b] Discuss IPV6 packet format. [4] CO3 L3

Q.5[a] Discuss and compare the three deployment models related to cloud computing. [5] CO2 L2 L3

[b] Enlist and discuss in brief the challenges related to IoT with cloud. [5] CO5 L3

Q.6 [a] Discuss the various criteria one should consider while selecting a cloud service provider. [5] CO5 L2 L3

[b] Discuss the various types of modelling algorithms used in machine learning. [5] CO5 L2 L3

Q.7[a] Discuss how IoT is used for smart warehouse monitoring by Amazon or Fedex. [5] CO6 L2 L3

[b] How IoT can be used for integrated vehicle health management? Discuss the end-to-end solution. [5] CO6 L2 L3

Q.8 Write short notes on any two. [2x 5] All CO's

i. Model building process in ML.
ii. Components of BLE
iii. XMPP
iv. Difference between edge and fog computing.
v. Service field in IPV4

Total no. of Pages: 03 *Roll no.....*

SEMESTER

B.Tech.

END TERM EXAMINATION ***NOV/DEC 2025***

HU-301 Engineering Economics

Time: 3:00 Hours ***Max. Marks: 50***

Note: All questions carry equal marks.
Answer any 5 questions.
Assume suitable missing data, if any.

Q.1 (a) What do you understand by Monetary Policy? Explain the tools of monetary policy to alleviate unemployment.

[5] [CO# 2and 4]

Q.1 (b) A company deposits ₹10,000 at the end of each year for 8 years in a fund that gives 14% interest compounded annually.
Find the future value of this fund at the end of 8 years. [5 Marks] [CO#3]

Q.2 (a) Discuss the salient features of the Indian economy. [5] [CO#1]

Q.2 (b) When the price of Product Y (petrol) rises from ₹80 per litre to ₹100 per litre, the quantity demanded of Product X (cars) falls from 500 units to 420 units. Calculate the Cross Price Elasticity of Demand (XED) and interpret the relationship between the two goods. [5] [CO#2]

Q.3 (a) Differentiate between Balance of Payment (BOP) and Balance of Trade (BOT). [5] [CO#4]

Q.3 (b) Complete the table: [5] [CO#4]

Output (units)	AVC (₹)	TC (₹)	MC (₹)
0	—	60	—
1	50	—	—
2	45	—	—
3	—	195	—
4	—	—	40
5	42	—	—

Q.4 (a) How does government expenditure contribute to reducing income inequality and promoting inclusive growth in India? [5] [CO#4]

Q.4 (b) **Zeta Ltd.** provides the following data for April: [5] [CO#3]

- Opening stock of Raw Materials = ₹1,20,000
- Raw Materials Purchased = ₹7,80,000
- Closing stock of Raw Materials = ₹1,00,000
- Direct Labour = ₹4,50,000
- Direct Expenses = ₹30,000
- Factory Overheads = 60% of Prime Cost
- Administrative Overheads = 10% of Factory Cost
- Selling & Distribution Overheads = 12% of Cost of Production

Assume opening finished goods = closing finished goods (so no adjustment).

Prepare a Cost Sheet showing only:

1. Prime Cost
2. Factory Cost
3. Cost of Production
4. Total Cost (i.e., Cost of Sales)

Q.5 (a) What is a business cycle? Explain the different phases of the Business Cycle. [2+3=5] [CO#1and2]

Q.5 (b) Alpha Industry is planning to expand its production operation. It has identified three different technologies for meeting the goal. The initial outlay and annual revenues with respect to each of the technologies are summarised in Table 4.1. Suggest the best technology to be implemented based on the present worth method of comparison, assuming a 20% interest rate, compounded annually. [5] [CO#3]

	Initial outlay (Rs.)	Annual revenue (Rs.)	Life (years)
Technology 1	12,00,000	4,00,000	10
Technology 2	20,00,000	6,00,000	10
Technology 3	18,00,000	5,00,000	10

Q.6 (a) How knowledge of engineering can be used to improve life at Slum?

[5] [CO#5]

Q.6 (b) A company invests in one of the two mutually exclusive alternatives. The life of both alternatives is 12 years, with the following investment, annual returns and salvage value. [5] [CO#3]

	Alternatives	
	A	B
Investment	8,30,000	7,45,000
Annual Equal Return	3,10,000	2,70,000
Salvage value	1,50,000	2,10,000

Determine the best alternative based on the annual equivalent method by assuming = 10%.

Q.7 (a) You are the CEO of a company planning to introduce a new product in the Indian market. What key factors will you analyse before launching the product? [5] [CO#5]

Q.7 (b) Consider the following data of a company for the year 1997: [5] [CO#3]

Sales = Rs. 1,20,000

Fixed cost = Rs. 25,000

Variable cost = Rs. 45,000

Find the following:

(a) Contribution

(b) Profit

(c) BEP

(d) M.S.